

Placement Preparation Portal (React + Spring Boot + MySQL) - Complete Learning Roadmap

Tech Stack

Frontend: React

Backend: Spring Boot

Database: MySQL

API Testing: Postman

Version Control: Git & GitHub

Overall Architecture

Frontend (React) → Spring Boot Backend → MySQL Database

MODULE 0: Setup & Fundamentals

Step 0.1 Install JDK 21, VS Code/IntelliJ, MySQL, Postman, Node.js, Git.

Learn: JDK, JVM, Node.js, Database, API.

Step 0.2 Create GitHub Repository. Learn Git, Repository, Commit, Push.

Step 0.3 Understand Full Stack Flow: User → React → Spring Boot → MySQL → Response.

MODULE 1: Spring Boot Backend Basics

Step 1.1 Create Spring Boot Project. Learn Maven, pom.xml, dependencies, project structure.

Step 1.2 Create First API (/hello). Learn Controller, REST API, GET Mapping.

Step 1.3 Learn Postman. Learn Request, Response, Status Codes.

Step 1.4 Create Student Entity (id, name, email, branch). Learn Entity, Table, Column, Primary Key.

Step 1.5 Connect MySQL. Learn Database connection.

Step 1.6 Create CRUD APIs: POST, GET, PUT, DELETE.

MODULE 2: Authentication System

Step 2.1 Create User Entity (id, name, email, password, role).

Step 2.2 Register API.

Step 2.3 Login API.

Step 2.4 JWT Authentication. Learn Tokens and Protected APIs.

MODULE 3: React Fundamentals

Step 3.1 Create React Project. Learn Components, JSX, Props, State.

Step 3.2 Build Home, Login, Register Pages. Learn Routing and Navigation.

Step 3.3 API Calling using Axios/Fetch.

MODULE 4: User Dashboard

Step 4.1 Dashboard UI: Aptitude, Coding, Notes, Progress.

Step 4.2 Protected Routes using JWT.

MODULE 5: Aptitude Module

Database table: aptitude_questions.

Fields: id, question, optionA, optionB, optionC, optionD, answer.

Question APIs and React Question Screen.

MODULE 6: Coding Questions Module

Fields: id, title, difficulty, description, sampleInput, sampleOutput.

Display Coding Questions using React.

MODULE 7: Notes Module

Fields: id, title, content, createdAt.

CRUD APIs and Notes UI.

MODULE 8: Progress Tracking

Fields: userId, aptitudeSolved, codingSolved, notesCreated.
Progress API and Dashboard Charts.

MODULE 9: Admin Dashboard

Admin Login.
Manage Questions.
Manage Users.

MODULE 10: Deployment

Backend Deployment: Render or Railway.
Frontend Deployment: Vercel or Netlify.

Final Database Tables

users, students, aptitude_questions, coding_questions, notes, progress.

Learning Format For Every Step

1. Concept
2. Real-Life Example
3. Diagram
4. Coding
5. Behind The Scenes
6. Interview Questions
7. Mini Task
8. Summary
9. Checkpoint (Done / Not Done / Error)